

Project Report: ZeroWaste

Bridging the gap between food surplus and food scarcity.

1. Project Overview & Problem Statement

The Global Crisis of Food Waste

Every single day, restaurants, bakeries, and grocery stores discard millions of pounds of perfectly edible food simply because it didn't sell before closing time. Simultaneously, local communities continue to face rising food costs and food insecurity. This creates a heartbreaking paradox: **we have an abundance of food, yet it ends up decomposing in landfills, generating harmful methane gases that accelerate climate change.**

The ZeroWaste Solution

ZeroWaste is a digital food rescue network built to intercept this waste. We provide a seamless platform that connects local food-based businesses directly with everyday consumers.

- **For Businesses:** It offers a streamlined, socially responsible way to offload daily surplus inventory—reducing disposal costs and building community goodwill.
 - **For Consumers:** It provides a real-time marketplace to claim high-quality, heavily discounted (or free) meals before they are thrown away.
 - **For the Environment:** Every meal rescued is a direct reduction in the carbon footprint associated with commercial food waste.
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2. Software Development Process (Agile & Sprints)

To manage the complexity of building a multi-role platform, our team utilized an **Agile Software Development Process** driven by iterative **Sprints**.

- **Iterative Planning:** We broke down the project into distinct Sprints. For example, *Sprint 1* focused exclusively on building the core foundation: Secure Authentication, the global React shell, and the Admin oversight dashboard.
 - **Role Allocation:** Features were divided among four team members, each taking ownership of specific modules (User Authentication, Marketplace Core, Claim/Verification Engine, and Platform Analytics).
 - **Continuous Integration:** By developing in Sprints, we could test and integrate independent modules (like the Login system and the Listing system) progressively, ensuring the app remained stable at the end of every cycle.
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3. Technology Stack (MERN)

To deliver a fast, responsive, and real-time experience, we architected ZeroWaste on the **MERN** stack:

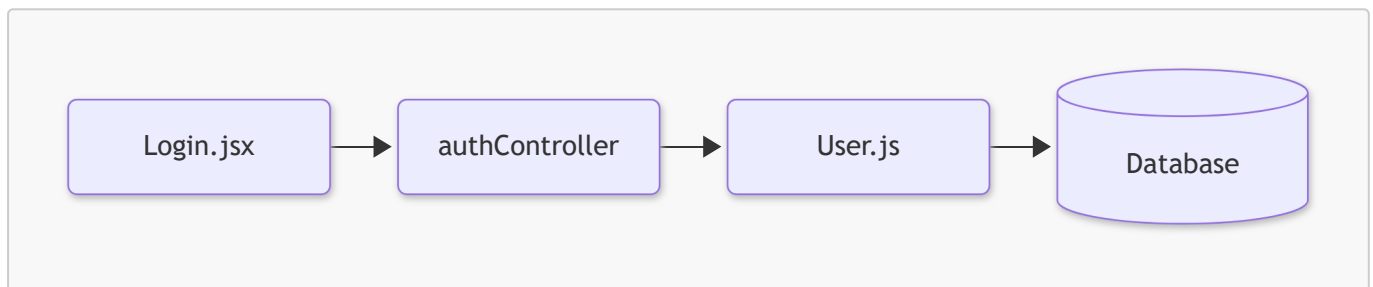
- **Frontend (React.js):** A dynamic, single-page application (SPA) providing an interactive UI.
- **Backend (Express.js & Node.js):** A lightweight, highly scalable RESTful API handling business logic.

- **Database (MongoDB & Mongoose):** A flexible NoSQL database perfectly suited for handling diverse data shapes.

4. MVC Architecture Implementation (Core Feature)

To keep our codebase organized, we adhered to the **Model-View-Controller (MVC)** architectural pattern. Here is the MVC breakdown for our most critical feature: **Role-Based Registration & Login**.

- **Model (User.js):** Defines the data blueprint. It enforces uniqueness for emails, dictates the user roles (Admin, Business, Consumer), and securely hashes passwords before they hit the database.
- **View (Login.jsx):** The presentation layer. A React component that captures user input, provides immediate visual validation, and sends the credential payload to the backend.
- **Controller (authController.js):** The brain of the operation. It receives the request, queries the Model to verify credentials, and if successful, signs and returns a secure JWT token back to the View.

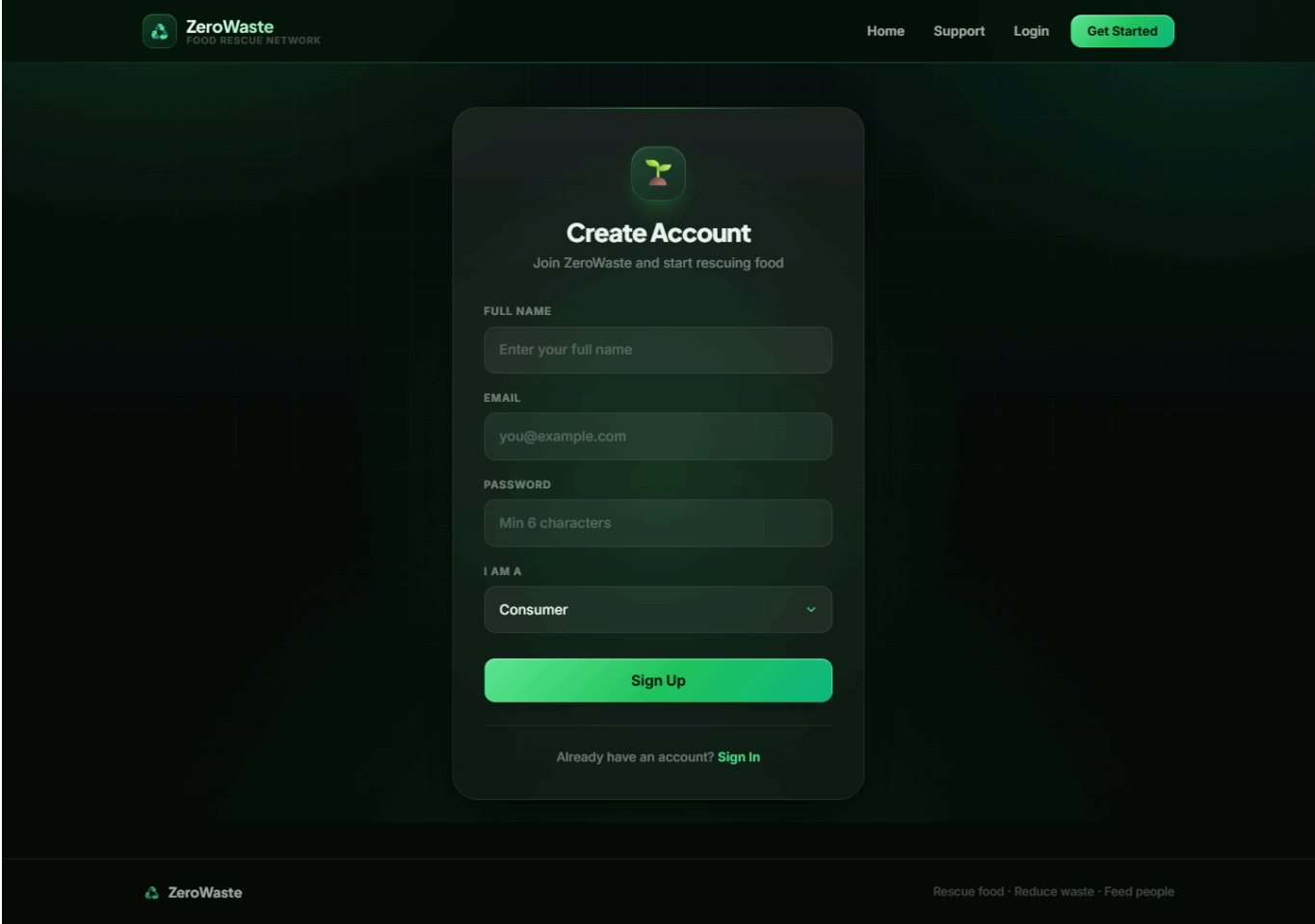


5. User Demonstration: 20 Core Features

Our platform offers a comprehensive suite of features tailored to our three distinct user roles. Below is a demonstration of the 20 features implemented across the application:

User Identity & Security

- 1. Role-Based Registration:** Users select their account type (Consumer, Business, Admin) upon signup.
- 2. Role-Based Login:** Users log in and are automatically routed to their specific dashboard based on their role.



3. Profile Management:

 Users can seamlessly update their personal details and contact information.

The screenshot displays the ZeroWaste user profile management interface. At the top, the ZeroWaste logo and navigation links (Home, Support, My Claims, My Pickups) are visible. The user is logged in as 'user01' (CONSUMER) and can click 'Logout'. The profile section shows a circular avatar with the letter 'U', the username 'user01', the role 'CONSUMER', and the email 'user@gmail.com'. Below this, the 'Edit Profile' section contains input fields for 'FULL NAME' (filled with 'user01'), 'EMAIL' (filled with 'user@gmail.com'), and 'PHONE NUMBER' (with a placeholder 'e.g. +880 1234 567890'). A green 'Save Changes' button is at the bottom of this section. The 'Change Password' section has input fields for 'CURRENT PASSWORD', 'NEW PASSWORD' (with a 'Min 6 characters' hint), and 'CONFIRM NEW PASSWORD'. A blue 'Change Password' button is at the bottom of this section. The footer includes the ZeroWaste logo and the tagline 'Rescue food · Reduce waste · Feed people'.

4. Password Security: Users can securely change their passwords, backed by bcrypt hashing. **5. Session Management:** The system automatically logs users out when their secure session (JWT) expires.

Business & Marketplace Features

6. Create Food Listing: Businesses can post surplus food with a title, quantity, and exact expiry time. **7. Image Uploading:** Businesses can attach real photos to their food listings. **8. Listing Management:**

Businesses can edit or delete their active listings from their dashboard.

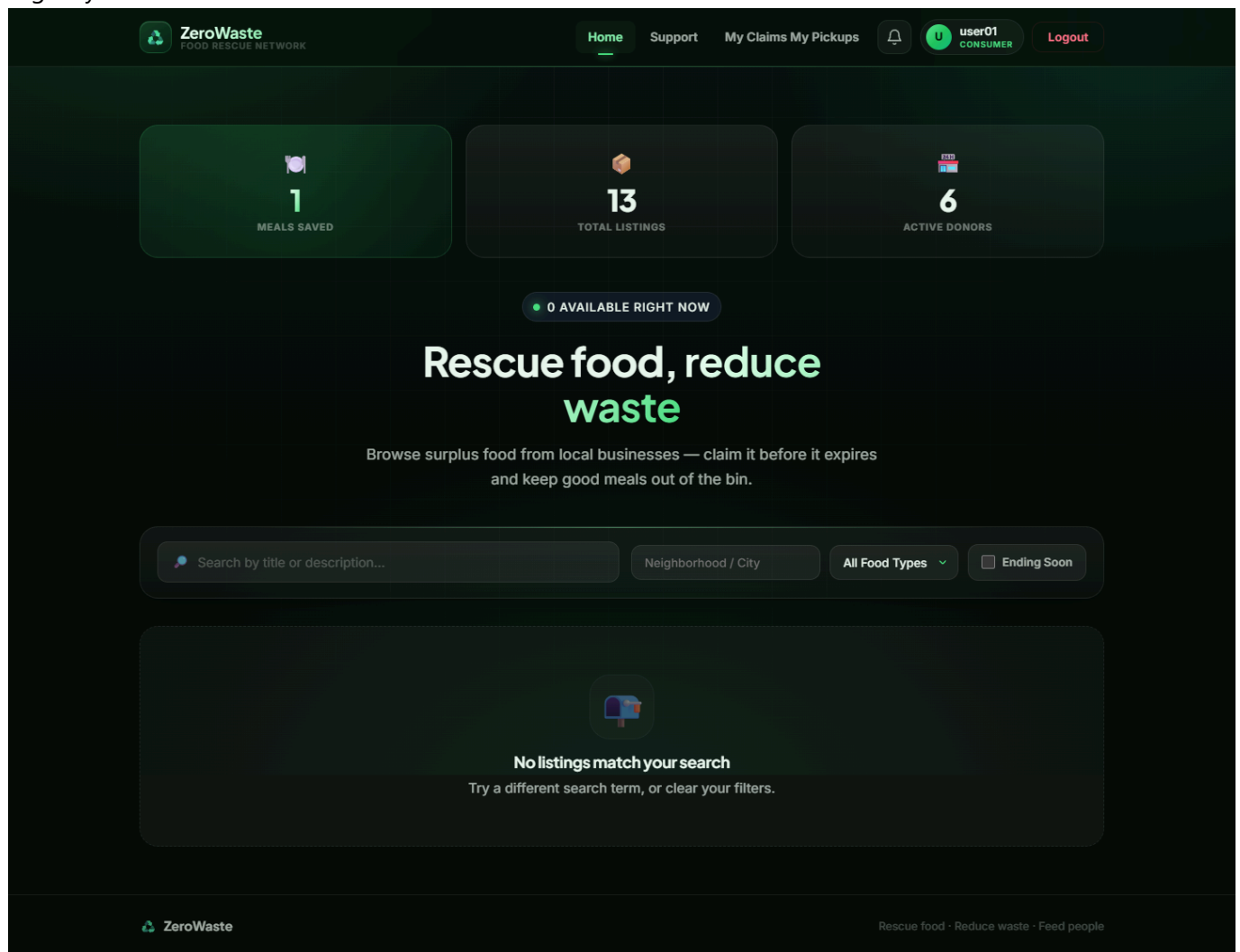
The screenshot shows the 'Listing Dashboard' for a business user on the ZeroWaste Food Rescue Network. The dashboard is dark-themed with green accents. At the top, there's a navigation bar with links for Home, Support, Listing Dashboard (active), a notification bell, a user profile icon labeled 'Test Cafe BUSINESS', and a Logout button. Below the navigation bar, the main content area is divided into two columns. The left column features a 'FEATURE 10' badge and a 'Listing Management' section with the text 'Create food listings with a photo for nearby users, and edit or remove them any time.' The right column displays three statistics: 0 LISTINGS CREATED, 0 ACTIVE LISTINGS, and 0 ITEMS POSTED. Below these, there's a 'NEW POST' section with a 'Photo Optional' toggle. The 'Create a listing' form includes fields for TITLE (e.g., 'Fresh bakery box'), DESCRIPTION (with a placeholder text), QUANTITY (12), FOOD TYPE (Veg), CITY (e.g., 'Dhaka'), NEIGHBORHOOD (e.g., 'Gulshan'), EXPIRY TIME (09/20/2026 01:09 PM), and a PHOTO upload section (currently showing 'No file chosen'). A green 'Publish Listing' button is at the bottom of the form. The right column also has a 'YOUR POSTS' section titled 'Recent listings' which currently shows 'No listings yet' and a prompt to 'Create your first post to start filling the marketplace.' The footer of the dashboard includes the ZeroWaste logo and the tagline 'Rescue food - Reduce waste - Feed people'.

9. Business Verification (OTP): Businesses securely verify consumer pickups by entering a 4-digit code provided by the consumer. **10. Business Analytics:** A dashboard for businesses to track their total donated items and completed orders.

Consumer Experience

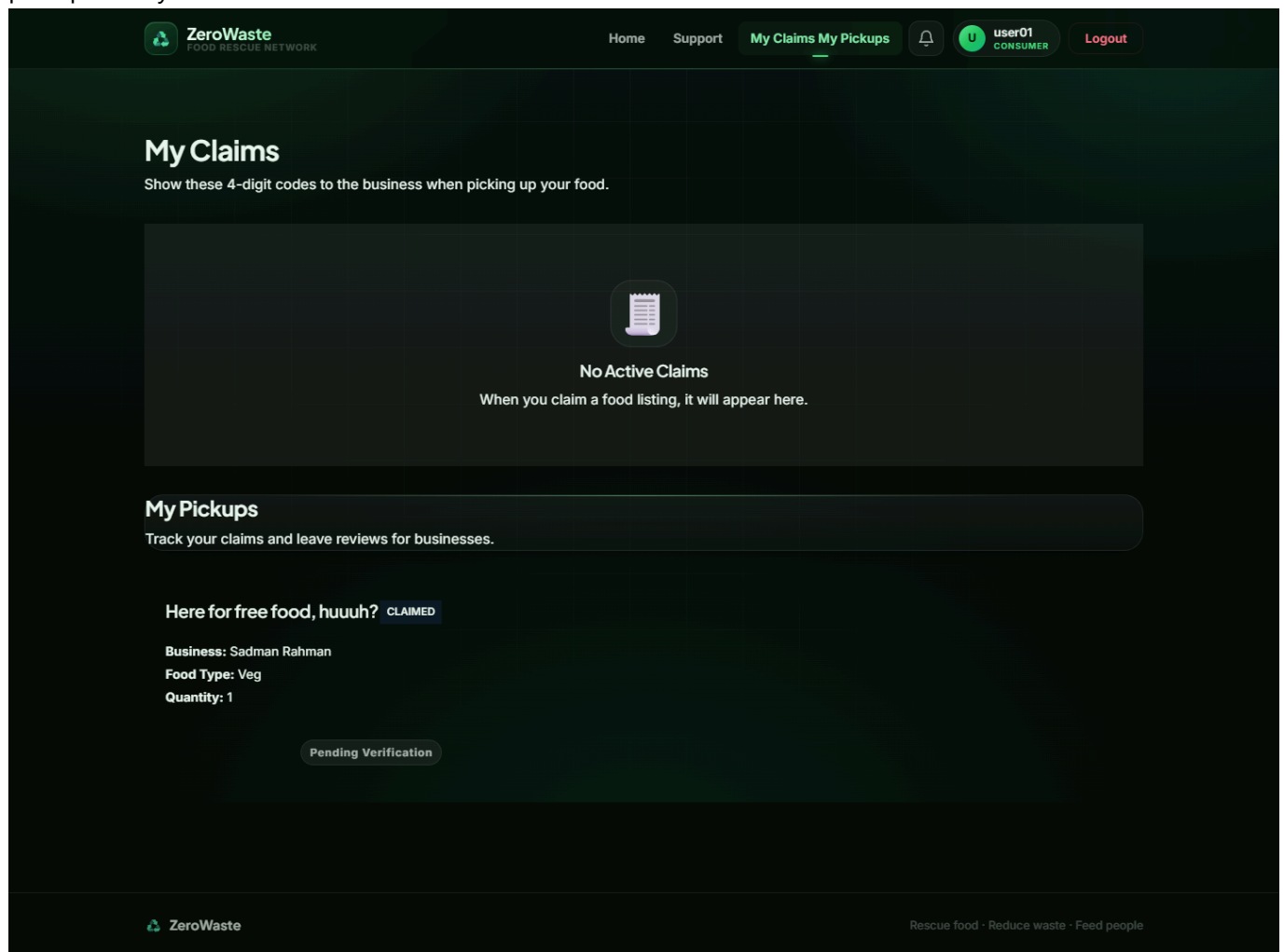
11. View & Search Feed: Consumers see a live feed of all active food listings with a dynamic search bar. **12. Filtering System:** Consumers can filter the marketplace feed by Neighborhood, Food Type (Veg/Non-Veg), or

urgency.



13. Claiming System: Consumers click 'Claim' to reserve food, instantly reducing the business's available inventory. **14. Pickup OTP Generation:** Upon claiming, consumers receive a unique 4-digit OTP to show the business. **15. Consumer Order Dashboard:** Consumers can view their pending claims and their historical

pickup history.



16. Review & Rating: Consumers can rate businesses out of 5 stars after a successful pickup.

Platform Administration & Impact

17. Admin User Management: A master view where Admins can see all registered accounts and ban or delete bad actors.

 **ZeroWaste**
FOOD RESCUE NETWORK

[Home](#) [Support](#) [Admin Panel](#)   **Admin Boss**
ADMIN [Logout](#)

Admin Dashboard

Manage users, moderation, and support tickets

[Users 18](#) [Support Tickets](#)

1
ADMINS

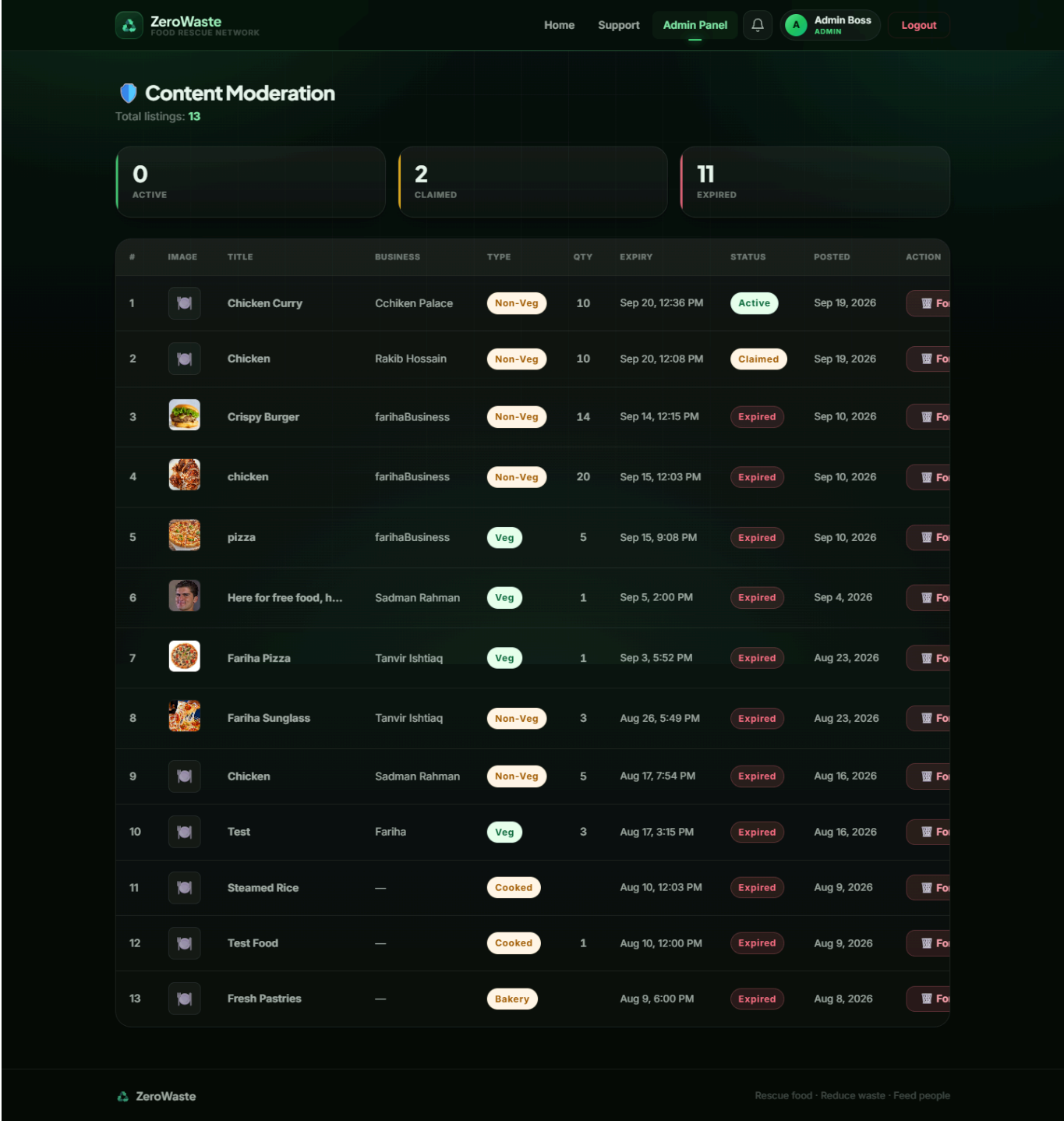
11
BUSINESSES

6
CONSUMERS

#	NAME	EMAIL	ROLE	JOINED
1	Cchiken Palace	palace@gmail.com	Business	Sep 19, 2026
2	Sadman Alfi	alfi@gmail.com	Consumer	Sep 19, 2026
3	Rakib Hossain	rakib@gmail.com	Business	Sep 19, 2026
4	FarihaUser	farihauser1@gmail.com	Consumer	Sep 10, 2026
5	Fardin Islam	nafiislam5688@gmail.com	Consumer	Aug 25, 2026
6	farihaBusiness	farihabusiness@gmail.com	Business	Aug 23, 2026
7	Tanvir Ishtiaq	tanvir@panoramamas.com	Business	Aug 23, 2026
8	Farihha Test 1	fariha.rahman3@g.bracu.ac.bd	Consumer	Aug 23, 2026
9	Other Biz	otherbiz.qa2@example.com	Business	Aug 23, 2026
10	Other Biz	otherbiz.qa@example.com	Business	Aug 23, 2026
11	Test Biz	testbiz.qa@example.com	Business	Aug 23, 2026
12	user01	user@gmail.com	Consumer	Aug 16, 2026
13	Fariha	fariharahman607@gmail.com	Business	Aug 16, 2026
14	Kacchi Bhaii	kacchibhai@gmail.com	Business	Aug 9, 2026
15	Sultan's Dine	sulatansdine@gmail.com	Consumer	Aug 9, 2026
16	Sadman Rahman	sadman.rahman@ac.bd	Business	Aug 9, 2026
17	Test Cafe	cafe@test.com	Business	Aug 8, 2026
18	Admin Boss	admin@zerowaste.com	Admin	Aug 8, 2026

 **ZeroWaste** Rescue food · Reduce waste · Feed people

18. Admin Content Moderation: Admins oversee all food listings globally and can forcefully remove inappropriate posts.



- 19. In-App Notifications:** A notification bell drops down to show real-time alerts (e.g., "Pickup successful").
- 20. Platform Impact Stats:** A public banner on the homepage dynamically displays the total number of meals saved across the entire platform.

6. Technical Challenges & Solutions

Building a multi-tenant platform presented several unique hurdles:

- 1. Complex Role-Based Routing:** We had to ensure three distinct user types (Admin, Business, and Consumer) were securely routed to their unique dashboards without cross-contamination. *Solution:* We implemented custom React Router guards paired with backend JWT middleware.
- 2. Database Relationships & Transaction Integrity:** Ensuring that when a Consumer "claims" a food listing, the inventory decreases accurately and an Order is generated without race conditions. *Solution:*

We designed a relational flow in our NoSQL database, tying the **Listing** ID to the **Order** model, and automatically generating the OTP.

7. Conclusion & Motivation

The core motivation behind **ZeroWaste** was the belief that technology should be leveraged to drive meaningful, tangible social change. By writing this code, we aren't just moving pixels on a screen—we are building tools that bridge the gap between excess and scarcity.

This project proved that with thoughtful software processes (Agile Sprints) and a dedicated team, we can build a platform capable of keeping thousands of pounds of food out of landfills and putting it on the tables of those who need it. ZeroWaste is more than just an academic exercise; it is a scalable blueprint for a more sustainable and compassionate future.